PARTS LIST

Semiconductors:	
IC1 (IC1A-IC1C)	- LM358 op-amp
T1	- 2N3904 NPN transistor
D1-D4	- 1N4001 rectifier diode
LED1, LED2	- 5mm LED
Resistors (all 1/4-watt, ±5% carbon):	
R1, R4	- 330-ohm
R2, R5	- 3.9-kilo-ohm
R3	- 22-ohm
VR1	- 4.7-kilo-ohm potentiometer
Capacitors:	
C1	- 1000µF, 25V electrolytic
C2	- 1F, 2.7V supercapacitor
C3	- 220µF, 25V electrolytic
Miscellaneous:	
CON1-CON3	- 2- pin connector
X1	- 230V AC primary to 6V-0-6V, 500mA secondary transformer
RL1	- 5V, 10A SPDT relay
S1	- SPST switch
S2	- Push-to-on switch
	- 230V AC device
	- Breadboard

of the mains supply. The voltage across supercapacitor C2 is applied to the inverting input pin 2 of IC1A, while the non-inverting input pin 3 is biased at a fixed voltage set by potentiometer VR1.

The voltage across potentiometer VR1 equals the voltage across LED2, which is identical to LED1. LEDs of any colour except blue or white can be used, as the forward voltage of blue and white LEDs exceeds the maximum voltage of supercapacitor C2.

The working of this circuit is simple. Initially, the voltage across supercapacitor C2 is low, making the voltage at the non-inverting input pin 3 of IC1A higher than at the inverting input pin 2. Consequently, the output at pin 1 of IC1A goes high, energising the relay through transistor T1. Conversely, when the voltage across supercapacitor C2 exceeds the voltage at the non-inverting input pin 3, the output of IC1A goes low, de-energising the relay. If the voltage across LED2 is approximately 2V, VR1 can be adjusted to provide 63.2% of this voltage (1.26V) to achieve a time delay of about one hour. Adjusting to about 1.73V results in a delay of two

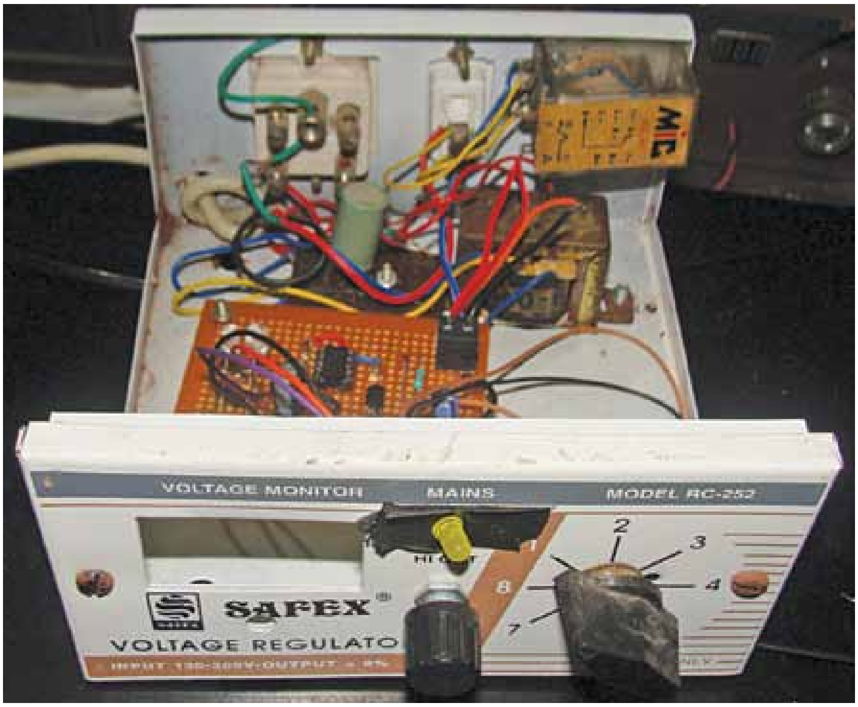


Fig. 3: Author's prototype

hours. Longer delays can be obtained by modifying the value of resistor R2. Since discharging a supercapacitor takes time, resetting the circuit immediately after it switches off can be done by pressing switch S2 for a few seconds.

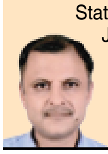
Construction and testing

To test the circuit in the EFY lab, it was assembled on a breadboard and housed in a suitable enclosure. After assembly, connect the mains power supply to the primary of transformer X1. The device to be controlled should be connected across connector CON2. This device will power on when switch S2 is pressed and will power off after the preset time set by VR1.

Switch S1 provides the mains power supply to the circuit, while switch S2 is used to reset or start the timer. Momentarily pressing S2 discharges C2, which then begins charging from zero. Potentiometer VR1 can be calibrated to produce a delay ranging from zero to several hours, depending on the values of R2 and C2. Diode D4 acts as a freewheeling diode to bypass the inductive spike of the relay coil.

Author's prototype is shown in Fig. 3, which is installed in a voltage stabiliser cabinet as it has space for fixing an AC outlet, transformer, relay and push to on switch.

EFY note. The circuit will work with any voltage up to 12V, and the time delay produced is independent of the input voltage. To obtain varying time delays, change the values of resistors R1, R4, and R5. The values of these resistors should be chosen so that R1 and R4 provide approximately 10mA to both LEDs, and R5 provides around 1mA base current. Ensure the use of a relay with an appropriate voltage rating to achieve the predicted time delays. EFY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